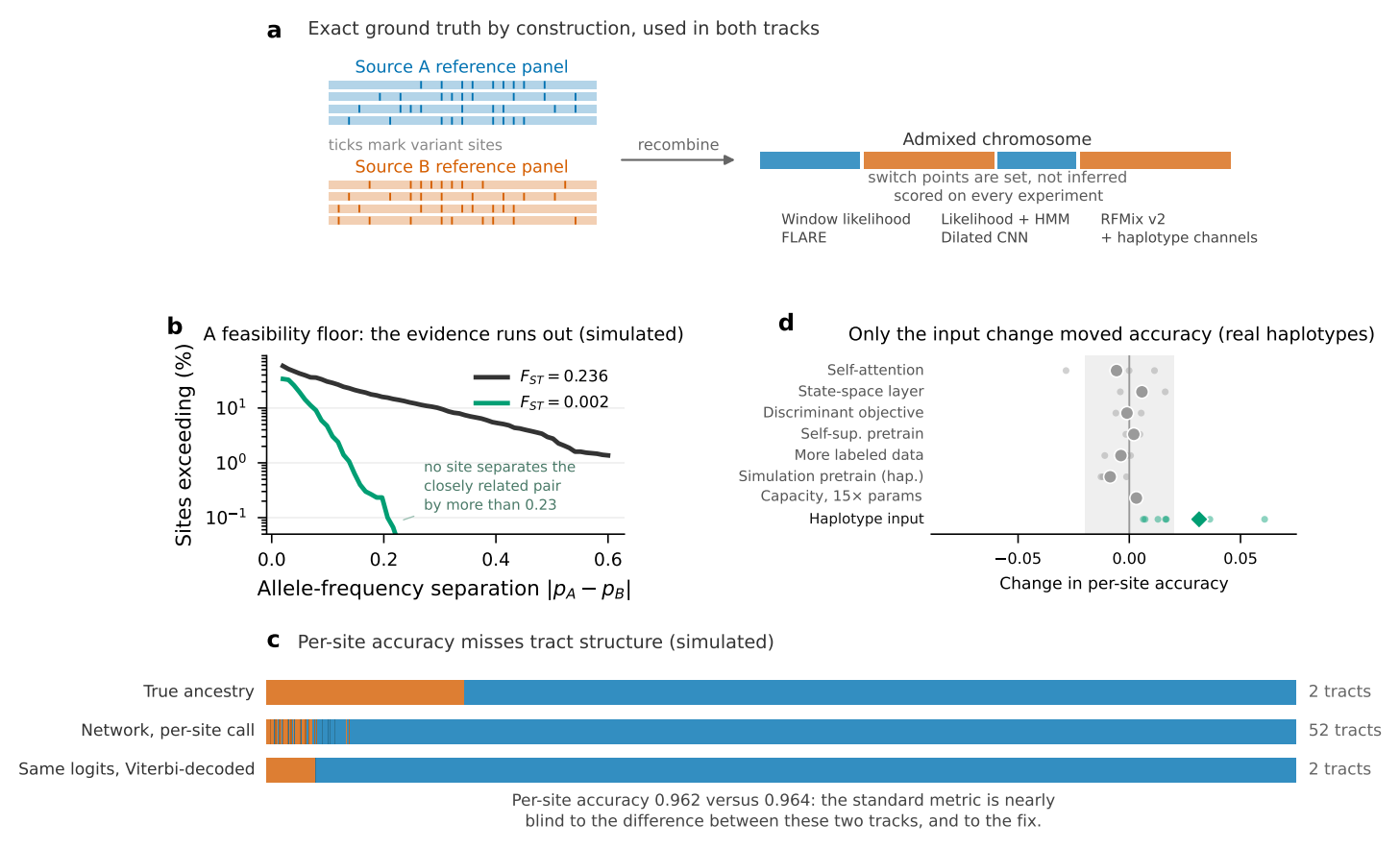
a
Exact ground truth by construction, used in both tracks
Source A reference panel
ticks mark variant sites
Source B reference panel
recombine
Admixed chromosome
switch points are set, not inferred
scored on every experiment
Window likelihood
FLARE
Likelihood + HMM
Dilated CNN
RFMix v2
+ haplotype channels
b
A feasibility floor: the evidence runs out (simulated)
Sites exceeding (%)
$F_{ST} = 0.236$
$F_{ST} = 0.002$
no site separates the closely related pair by more than 0.23
Allele-frequency separation $|p_A - p_B|$
d
Only the input change moved accuracy (real haplotypes)
Self-attention
State-space layer
Discriminant objective
Self-sup. pretrain
More labeled data
Simulation pretrain (hap.)
Capacity, 15× params
Haplotype input
Change in per-site accuracy
c
Per-site accuracy misses tract structure (simulated)
True ancestry
Network, per-site call
Same logits, Viterbi-decoded
2 tracts
52 tracts
2 tracts
Per-site accuracy 0.962 versus 0.964: the standard metric is nearly
blind to the difference between these two tracks, and to the fix.